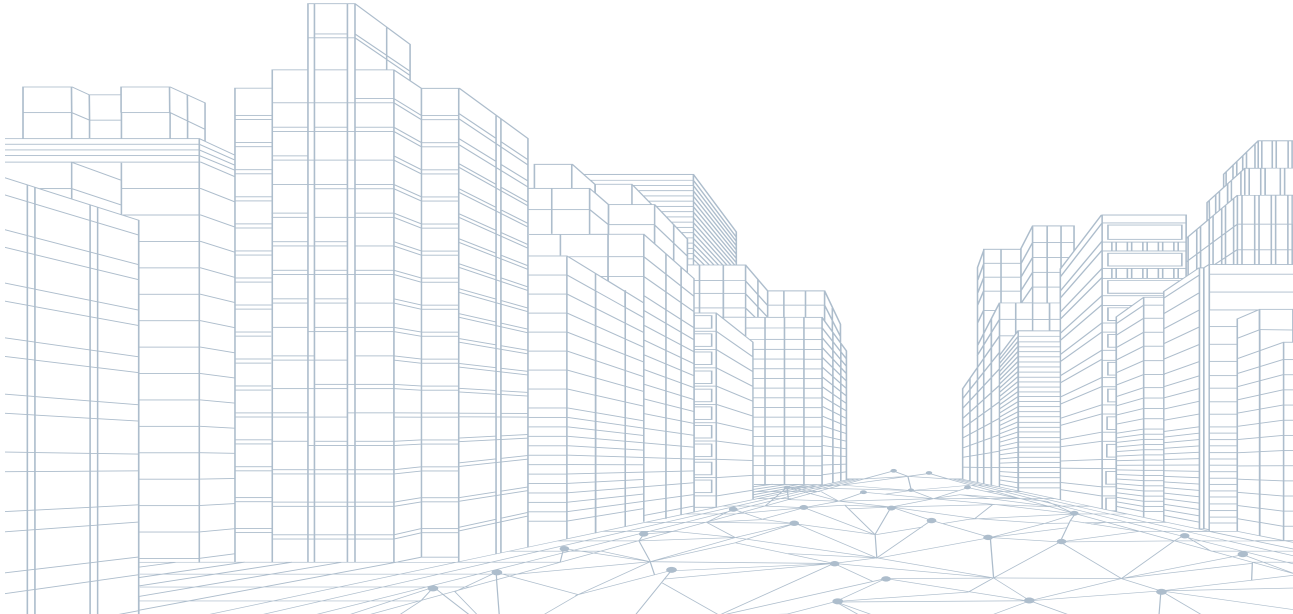
Lecture 1
Introduction to Computer Networks

IT5100D Industry Readiness: Networking Concepts for Developers

AY2025/2026 Semester 2

Vivy Suhendra



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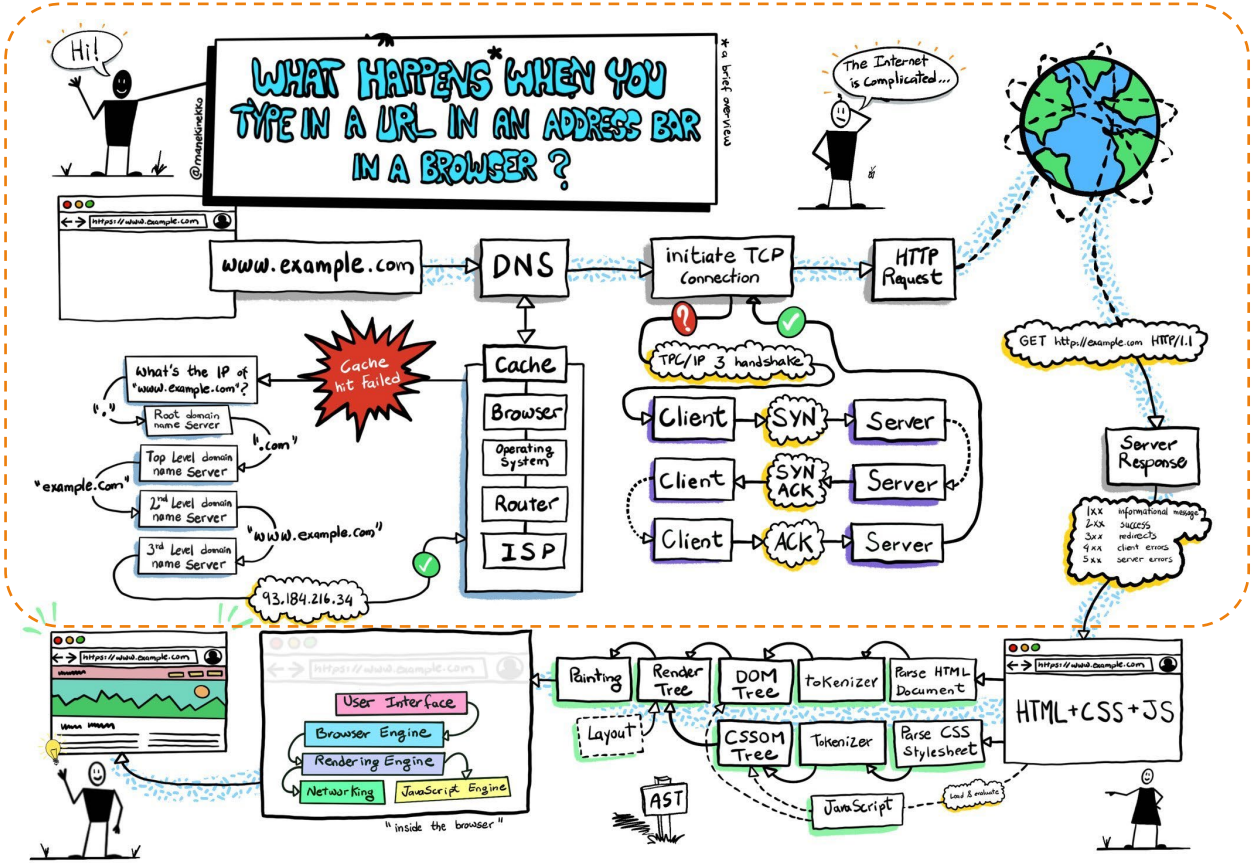
Age-Old (CS) Interview Question?



A big part of this course is to understand this portion

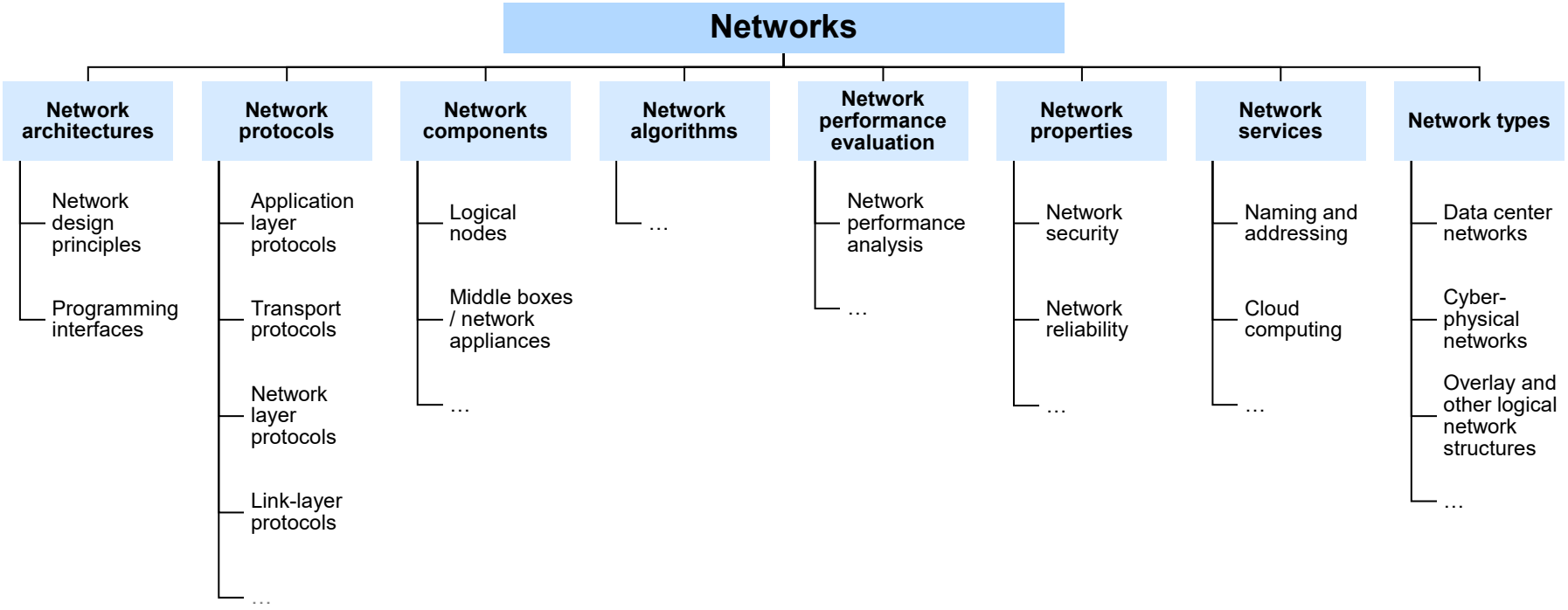
Credit: [@manekinekko](#) 2020

Just for fun: [Super-detailed explanation](#)
(70+ people collaborative effort)

Computer Networking is a Vast Field

Subset of the 2012 ACM Computing Classification System

(lower levels not shown)



Why Networking Is Complex

- Computer networking is an active and rapidly changing field.
- Many technologies exist, each with distinguishing features.
- Technologies can be combined and interconnected in many ways.
- No single underlying theory that explains the relationship among all parts.
- Multiple networking standards, some incompatible with others.

-- Douglas Comer, "Computer Networks and Internets" (6th ed., 2015)

Our Goal in this Course

- Understand (at a high level)
  - How computer networks work
  - How networked applications work
- Be aware of networking factors that affect web applications
  - Reliability, speed, other performance metrics
  - Security

- This course focuses on the network, <u>not</u> the web application aspect
  - IT5007 → web development / deployment / best practices
  - Relevant: IT5100E → dedicated to security best practices for web application

- This course anchors on “stable” concepts rather than specific technologies
  - Concept understanding will help you learn new protocols, techs that will keep coming up

Student Background Survey

- Statistics on Canvas

- What do you hope to learn in this course?
  - **Gain / deepen** understanding of networking that will be useful for work/interview
    - How network protocols / architectures / security mechanisms work in real-world systems
    - Networking best practices
    - Knowledge useful for debugging/troubleshooting, e.g., how to test connectivity
  - Things more related to web application (?)
    - Developing and deployment
    - Backend web dev knowledge

Lesson Plan

WEEK	TOPIC	ASSESSMENT
7	Introduction to Computer Networks	
8	Network Applications	Assignment 1 (**20%**) – due in 2 weeks
9	Network Addressing	
10	Network Performance *Mock Test with Wireshark*	Assignment 2 (**20%**) – due in 2 weeks Mock Test (**5%**) – due in 3 weeks
11	Network Security	
12	Networking Technologies: Internet of Things	Assignment 3 (**25%**) – due in 2 weeks
13	*Final Test*	**FINAL TEST (30%)**

Assignments

- Practical exercises to complement theory from lecture
- Always submit your own work only.
- Plagiarism penalty (submitting identical proof of work): 0 mark for the question and –50% of total mark
- Late penalty: –20% of total mark every 12 hours

Use of AI Tools in Assignments

- You can use AI tools for ideas, but be responsible for your own learning and work.
  - Use only what you understand and test / verify the correctness.
  - Acknowledge any use of AI tools in what you submit (see sample on submission page).
- NUS view: representing an AI's output as your own work is plagiarism.
  - Read AI Guidelines for Students

Final Test

- In-person during Week 13 lecture, **13 Apr 2026**
- NO remote arrangement. Only medical or reservist reasons accepted for make-up.

Test Mode

- To be taken on Examplify using your own laptops
- Open book with no internet (Non-Secure Block Internet) – iPad is not supported
- MCQ + short answers + essay, some Wireshark trace file analysis

Use of AI Tools in the Test: not allowed

- Any local Large Language Models (LLMs) installed must be completely removed before the test.
  - Read the NUS policy: MyPortal @ NUS > Instructions to Candidates for Assessments and Examinations > Ban on Local Large Language Models

Violation of exam rules → disciplinary action (usually leads to F grade)

Mock Test

- Exercise on Wireshark analysis, done in Examplify
  - Familiarize with Final Test setup
  - Check any issue with Examplify + Wireshark on your laptop
- Focus only on Wireshark trace file analysis, not full scope of the test

- We will run this in class during Week 10 lecture
- Or you can complete it on your own time and submit within 2 weeks

- More details nearer the time

Addressing Past Course Feedback

Assignment

- Provide more help on assignments, they were not trivial for someone new to networking
- Appreciate the depth, but the volume is too much

→ Added Wireshark tutorial to help with tasks in assignments

→ Trimming down assignments, moving some tasks to in-class exercise / extra resource

Test

- More practice questions to help with the test
- Materials tested are not covered in lectures

→ Adding mock test; including more past test questions in quizzes

→ Clarification: Test may also cover knowledge from doing assignments; deeper understanding of concepts

Wireshark

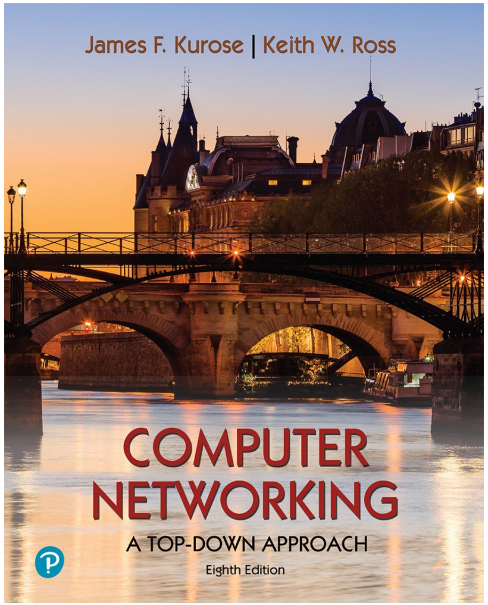
- Working with Wireshark is interesting and useful in industry, good to provide more guidance if possible

→ Including more demos in lecture; added Wireshark tutorial

Credits

Some illustrations taken from the resources of this book:

Computer Networking: A Top-Down Approach
Jim Kurose, Keith Ross



Today we will learn:

- What a **computer network** is
- How **the Internet** is structured
- The concepts of **protocols**
- **Layered** internet protocol stack

Q&A

- Raise your hand
- PollEv: https://pollev.com/vivy

- Outside class hours: Canvas Discussion